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Human Capital Theory and Higher Education in Developing Countries

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Introduction

The selection of human capital theory as a lens for evaluating the development of higher education comes not from specific advocacy by the author, but rather from considerations of practicality. It is arguable that any proposal to introduce or renovate higher education models in developing countries without factoring in human capital theory will languish. In this discussion of the human capital theory of education, five main questions are addressed: (a) why is it important; (b) what is it; (c) how has it affected higher education reform in developing countries; (d) is the scope of this theory sufficiently inclusive to rationalize fully models for higher education reform in developing countries, and (e) might there be an alternative yet related theory that should be considered? This literature review initially laid essential theoretical groundwork for a research study on higher education in Vietnam as a case of one developing country; therefore, some aspects of this examination into human capital theory are specific to Vietnam's context.

Importance of the Theory

The human capital theory of education derives its significant influence from the powerful company that it keeps. Among the proponents of this education theory are the World Bank, Organization of Economic Coopera-

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tion and Development (OECD), United Nations, European Union, United States, United Kingdom, Japan, and Singapore (Spring, 1998). Organizations that have been particularly active in promoting the human capital theory of education to developing countries are the World Bank, International Monetary Fund (IMF), and OECD (Karabel & Halsey, 1977).

The argument that developing countries must be integrated into the global economy has resulted in a new perspective concerning the relationship between education and development (Ayres, 2000). The World Bank, "as the largest supplier of external finance to education, has been instrumental in promoting this new perspective" (Ayers, 2000, p. 445). Spring (1998) states that, "in recent years, the World Bank has made education loans a top priority" (p. 179). Human capital theory "is evident in the educational policies of those Third-World nation-states influenced by the World Bank . . . which embrace among their foremost priorities the universally applied notion of 'human resources development'" (Ayres, 2000, p. 446).

Because developing countries are fighting wars against poverty, economic development is absolutely critical to their success. The appeal of human capital education theory comes from the donor funding that accompanies it, but perhaps also from its promise to support economic growth through the one resource in which these countries have confidence, the latent talents and intellectual abilities of their people.

Content of the Theory

Spring (1998) traces the beginning of human capital theory back to the 19th and early 20th centuries, when many in the United States were convinced that the corporate model of education would provide external efficiency in relation to the labor market and, at the same time, ensure equal opportunity. "The school functioned as a sorting machine to separate and train human resources to meet the demands of the labor market" (Spring, 1998, p. ix). The corporate model evolved into a global model and was accelerated when neocapitalism became the popular paradigm in the 1960s. Unlike laissez-faire economics, "neocapitalism calls for government intervention to promote and protect free markets. . . [and] education should be evaluated as an economic investment" (Spring, 1998, p. x).

One of the best known proponents of human capital theory is Theodore W. Schultz, an economist and Nobel prize winner. Schultz (1961) argued that, "measured by what labor contributes to output, the productive capacity of human beings is vastly larger than all other forms of wealth taken together" (p. 313). In Schultz's (1961) view, laborers are

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capitalists because the acquisition of "knowledge and skill are in great part the product of investment and, combined with other human investment, predominantly account for the productive superiority of technically advanced countries" (p. 314). According to Little (1999), Schultz's concepts were particularly influential in the 1960s and 1970s among those who worked in development and those who advocated the growth of formal education internationally: "The production of an educated labor force was perceived by both economists and development planners as a means to an end of the growth of the national economy, and hence, development" (p. 17).

Fundamentally, "human capital is a concept based on the belief that the role of workers in production is similar to the role of machinery and other forces of production" (Johnson, 1995, p. 133). Thus the purpose of investing in schools is to provide essential knowledge, skills, and abilities for workers and is evaluated in the same way as investment in facilities and equipment for the purpose of improving productivity. OECD defines human capital as the knowledge acquired throughout life and applied to production of goods, services, and ideas in a variety of contexts: "This definition allows accountants to apply the same techniques to human capital as are used for the potential output of a machine" (Spring, 1998, p. 169). Thus human worth is measured against various production criteria.

Of course, many people might find the concept of investment in human capital to be objectionable: "Our values and beliefs inhibit us from looking upon human beings as capital goods, except in slavery, and this we abhor" (Schultz, 1961, p. 313). Nevertheless, according to Karabel and Halsey (1977), human capital theory has a "direct appeal to pro-capitalist ideological sentiment that resides in its insistence that the worker is a *holder of capital* (as embodied in his skills and knowledge) and that he has the capacity to invest (in himself)" (p. 13). Put in a more positive light, Schultz argues that "by investing in themselves, people can enlarge the range of choices available to them. It is one way free men can enhance their welfare" (p. 314).

According to Ashton and Green (1996), "the fundamental weakness of the theory . . . is that in regarding human capital as a 'thing' to be acquired or utilised alongside other factors, it misses the social context of skill and of technology" (p. 17). They suggest a different name for the theory, calling it "the 'theory of capital humans' as this would capture the fact that it is humans whose skills are being objectified, rather than physical capital which is being dignified with humanity" (p. 18). They are not claiming that the theory is false, but it treats education and training as a "black box" production process that is the educators' responsibility and this process is most often not informed by the economists and

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business community (Ashton & Green, 1996, p. 18). Another problem associated with the methodology of human capital theory is that human beings can be pragmatic: "Individuals make endogenous choices about their education, reflecting differences in developed ability or personal tastes" (Belfield, 2000, p. 32).

OECD has consistently worked to develop an effective human capital accounting system. One model for analyzing the monetary value of human capital begins with the decision to acquire certain knowledge for the purpose of increasing production or expansion (Spring, 1998). The inventory of current personnel is reviewed to determine if any have the required skills and whether it is most cost effective to train employees or hire new people. Trends in these decisions can affect government policies and future human capital investment decisions. According to Fizsimons (1999), "under Human Capital Theory the basis for nation state structural policy frameworks is the enhancement of *labour flexibility* through regulatory reform in the labour market, as well as raising skill levels by additional investment in education, training and employment schemes" (p. 1). Yet Spring (1998) contends that "despite the arguments of human capital accountants, OECD policy experts continue to be frustrated by their inability to assess the economic return on social investment in education" (p. 171).

Toward the end of the 1960s discussions arose concerning the validity of some human capital theory assumptions, "particularly that higher earnings reflect greater marginal productivity and that national differences in rates of economic growth can be causally attributed to varying degrees of educational investment" (Karabel & Halsey, 1977, p. 308). The methods of cost-benefit analysis, using purely economic concepts, had also become controversial "since cost-benefit analysis has not succeeded in measuring all the indirect benefits of education, or other types of social investment (Woodhall, 1987, p. 3). Therefore, human capital theory did not provide a complete answer to the questions associated with resource allocation.

Another significant problem with basing government education policy and resource allocation decisions on human capital theory is the fallibility of labor market, often called "manpower," forecasting. According to Woodhall (1987), many economists criticize the forecast method because it is based upon three flawed assumptions. First, there is no fixed relationship between the educational qualification of the laborers and industrial output, as is assumed in the forecasts. Second, "there are no fixed educational requirements for the majority of jobs" (Woodhall, 1987, p. 5). Finally, technological change makes long range forecasting of acceptable accuracy nearly impossible.

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The value of human capital theory has been criticized based upon the contention that one of its fundamental assumptions is incorrect. The assumption is "that the perfect competition prevailing in labor markets ensures that greater earnings reflect greater productivity" (Karabel & Halsey, 1977, p. 14). Thus human capital theory is weakened by its dependency on wage theory. Thurow (1977) states that "the acceptance of the efficacy of education is itself derived from a belief in the standard economic theory . . . [that] the labor market exists to match labor demand with labor supply" (p. 325). Yet he argues that there is evidence contradicting the wage theory approach. For example, Thurow says that in the United States the labor market is driven by job competition rather than wage competition: "Instead of people looking for jobs, there are jobs looking for people — for 'suitable' people" (p. 326). When job competition exists, the function of education is to certify that the workers are trainable and to provide them with a certain status based on this certification (Thurow, 1977). Additionally, Karabel and Halsey (1977) point out that wages are influenced by numerous factors including minimum wage regulations, customs, and status.

To summarize the discussion of human capital theory thus far, it treats human knowledge, skills, and abilities as production tools and education becomes a capital investment decision. Two weaknesses inherent in this theory are its dependency on wage theory, which is flawed, and the difficulty of measuring education outputs using economic criteria. Yet to fully evaluate human capital theory in education, it is necessary to examine some of the consequences that result from the theory's implementation.

Consequences of the Theory

As a proponent of human capital theory for education, the World Bank defines basic education in terms of subjects that promote economic growth, thus leaving out such fields of study as history and literature (Spring, 1998). Yet Johnson (1995) states that "in every education system SOCIALIZATION ensures that new members know how to participate . . . . Schools are the social institutions in which education takes place" (p. 92). It would then seem that the human capital education model produces a society entirely focused on economic production, although some would say that educated humans also become more sophisticated consumers (Schultz, 1961).

The World Bank's concept of external efficiency imposes human capital theory on developing countries through the establishment of economic goals. Spring (1998) explains that "external efficiency is deter-

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mined by the number of unemployed college graduates. The higher the number of unemployed graduates, the lower the efficiency of the higher education system" (p. 183). Therefore, according to human capital theory, the goal of becoming educated is to find a job. With regard to education and the market, the World Bank's position is that "you can force potential students to make better labor market decisions by imposing or increasing tuition. In addition a private higher education system can be cultivated and public institutions can be forced to rely on private financing" (Spring, 1998, p. 183). Fewer students can afford the tuition, which reduces the number of graduates. According to the theory, decisions made by the students concerning future employment opportunities exert pressure on higher education institutions to alter their programs and curricula. Since private institutions must be competitive, they are sensitive to the needs of the labor market.

In summary, implementing human capital goals restricts access for higher education admission and results in less financial support, hence fewer offerings of subjects that do not link to economic measures, including art, literature, and history. Since labor market forecasting is recognized as being quite difficult, the simple answer to unemployment is to raise tuition and lower the number of graduates. Not considered in this theory are some essential critical shortage labor areas that do not receive incentives from the economy, specifically teachers and social workers.

Elements Missing from the Theory

In relation to citizenship and socialization, education is considered by many to be the foundation of democracy, which requires participation by citizens who can analyze, dialogue, and support or lead constructive social initiatives (Gay, 1994). In Vietnam, where the political and socioeconomic structures are quite different from those found in the U.S., institutions of higher education are required to teach good citizenship and all students must take 250 hours of Marxism-Leninism (Tran, 1999), which includes the lessons of Ho Chi Minh. According to MOET (2000), "tertiary education makes [a] contribution to the development of science-technology, culture-art, production, business and service simultaneously helping knowledge improvement, [and] healthy life style building among [the] people strata" (p. 44).

Other objectives for higher education are represented in the New Jersey Commission on Education (1999) goal statements: "The state's higher education system will develop and nurture the citizens and leaders of the future, preparing individuals for fulfilling lives, rewarding careers, and lifelong learning" (p. 3). The term "lifelong learning"

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associated with this context is much broader than the conceptual definition used by human capital theory, in which "lifelong learning as a social investment is supposed to reduce apprehension about constant social change and dislocation resulting from market competition and new technology" (Spring, 1998, p. 167). As noted previously, there are also intrinsic factors that motivate people to become more educated or trained in certain skills. A second New Jersey goal reflects the functions assigned to higher education institutions: "Through teaching, research, and public service, colleges and universities will support the state's public policy goals for economic growth, social stability, and enhanced quality of life" (New Jersey's Commission on Higher Education, 1999, p. 3).

Edward Shils (1982) thought that the primary task of universities "is the methodological discovery of and the teaching of truths about serious and important things" (p. 3). He argued that "part of the task is to enhance the students' understanding and to train them in the attitudes and methods of critical assessment and testing of their beliefs" (p. 3). Thus students develop into analytical and critical thinkers. According to Bok (1990), the most important function of higher education is to serve society: "Advanced training, specialized knowledge, and scientific discovery are now essential to solving many urgent problems facing our civilization — problems of disease and health, of the environment, of economic progress, of human survival" (p. 11). Bok's concept fits well with Clark's (1983) description of academics across nations who see their lives as a commitment "to serving knowledge, youth, and the general welfare, avoiding the crass materialism of the marketplace" (p. 74).

Bok (1990) argues that moral education is important to society and universities can make a positive contribution in this area. He points to courses in applied ethics that require background reading on ethical theory and discussion concerning practical application in personal and professional situations. The primary purpose of these courses is to help students be "more perceptive in detecting ethical problems when they arise, better acquainted with the best moral thought that has accumulated through the ages, and more equipped to reason about the ethical issues they will face" (Bok, 1990, p. 73).

For Vietnam, an important consideration is the potential effect that Confucian philosophy has had on society and education. Although Confucian influence declined during the 20th century under French colonial rule, "its basic concepts remain deeply imbedded in Vietnamese morality and values" (Pham, 1994, p. 125). The primary goal in Confucianism is the pursuit of virtue, which includes self-discipline, family cohesiveness, morality, and scholarly learning. In Vietnam, one of the five obligations was Tri, the duty to become educated (Pham, 1994). For

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Confucius, "knowledge was held as an indispensable treasure. The knowledge which he taught to be pursued was not purely scientific learning, but was the study of the sacred texts and the rules of virtue and propriety" (Shinn, 1997, p. 3). As a result of this Confucian influence, education is highly valued in many Asian countries.

Confucian philosophy, as well as the concepts of education stated by Gay (1994), Shils (1982), and Bok (1990), conflict with human capital theory at two significant points. First, under human capital theory, social investments must demonstrate economic benefits; therefore, moral and social obligations are not the primary considerations. By contrast, "the Confucian emphasis on filial piety makes it a moral obligation for the young to ensure that the aged receive meaningful and rewarding retirement" (Spring, 1998, p. 164). In the education context, most Vietnamese universities, through their students and staff, perform community services. Clearly, Bok and Gay think that education institutions have strong social obligations that go beyond economic linkages.

Second, under human capital theory, the value of knowledge is determined by the extent to which it promotes economic growth. Confucius supported the idea of tests to measure and select wise leadership, but the focus was on identifying those scholars who were capable of creating a moral and just society (Spring, 1998). Bok (1990) supports teaching students applied ethics, and Shils (1982) emphasizes the importance of faculty ethics in teaching students how to think critically and to seek truth. In summary, education has multiple essential missions, and human capital theory only focuses on one of them to the exclusion of the rest.

The Theory as it Plays in Vietnam

Vietnam's higher education system shows that World Bank recommendations have been incorporated into the "renovation" process; student fees have been implemented, the institutions are diversified to include private and semiprivate higher education, and access controls are exercised to channel students into fields of study that support economic needs. Vietnam seems to be making progress in developing its higher education system, but the emphasis on human capital theory may be producing mixed results. For example, Tran (1999) says the market economy has changed higher education in Vietnam, "but alongside the positive aspects of this transformation, 'money mania' and corruption — the negative features of unfettered capitalism — have also appeared" (p. 2).

Another significant point made by Tran (1999) concerns the contradiction between Vietnam's economic plan and its form of governance:

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"Vietnam is probably the only country in the world that embraces market economics and adheres to Marxism-Leninism. How can an education system guided by two contradictory philosophies develop in a consistent manner" (p. 2)? The answer to this question may be in the control mechanisms associated with human capital theory; MOET closely manages accession numbers into specific fields of study. Access to various fields is also influenced by the allocation of government scholarships.

An Alternative Concept

In discussing the history of human capital theory, Little (1999) says that the influence of this theory declined in the 1970s because the economic benefits expected from education either did not manifest themselves or they came forth too slowly. Then, in the 1980s and 1990s, there was a resurgence in the concept of education as a primary role player in development. Governments were encouraged to invest in education as a means for increasing their economic competitiveness: "[Education] was also seen as a vital ingredient in meeting global, as distinct from national challenges, such as worldwide economic stagnation, environmental degradation, civil war, and population growth" (Little, 1999, p. 18). This renewed emphasis on education used human capital theory, but there was an important, although perhaps subtle, difference: "Economic growth began to be posited as a means to the end of human development, rather than an end in itself" (Little, 1999, p. 18).

The human development concept, as used in relation to national and international projects, "emphasized the growth of human potential as well as the growth of income, the quality of life, the participation of people in their own development, and human freedom" (Little, 1999, p. 18). Education thus conceptualized was increased in breadth to include quality of learning, the experiential aspects of learning, and greater equity for those who had been previously excluded, such as girls, ethnic minorities, and remote populations.

Foster (1987) adds another significant perspective in analyzing the interactive relationship between education and economic development: "Educational expansion is initially consequent upon a degree of structural economic change having already occurred in the local economies, though it may in turn become a catalyst for further economic development" (p. 94). Because economic development is not a smooth linear process, the importance of education is variable over time. Human capital theory has failed to account for this dynamic in its cost-benefit calculations and forecasts. Foster concludes that "in fact, rather than constituting the key to economic and social development, education

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might be properly regarded as a component in a jigsaw puzzle whose linkage with other pieces is as yet obscure" (p. 94).

The human development concept seems more acceptable than human capital theory as a basis for formulating education models in developing countries because it offers greater flexibility and inclusiveness in relation to the goals of education as well as the corresponding needs of society. By thinking in terms of human development one can still include the important, perhaps even leading, role that education plays in economic development, without excluding other significant functions.

Conclusion

The contradictions found in Vietnam may actually be consistent with a basic contradiction found in human capital theory. According to this theory, the objective of education and training is to find a job, so students will choose institutions that have relevant programs in labor market shortage areas. Competition among institutions will then help shape the higher education system, especially in the case of private institutions. On the other hand, cost-benefit analysis and labor market forecasting help the government make informed management decisions and policies that control the higher education system programs and student enrollments through resource allocation and manipulation of student fees. This means that the hidden hand of competition is supposed to operate alongside centralized government controls. However, another problem with human capital theory is that in developing countries, students compete to gain access to higher education, the institutions do not compete for students: "Resources, both public and private, have not kept pace with escalating enrollments and costs" (Ransom, Khoo, & Selvaratnam, 1993, p. 1).

Because organizations like the World Bank, IMF, and OECD use the human capital theory of education, developing countries cannot avoid referring to this theory as a structural guideline for higher education reform. Yet human capital theory has significant weaknesses, and these countries cannot afford to exclude the other functions served by education. Perhaps the most promising approach to resolving this problem is found in Little's (1999) discussion of the human development concept. From this perspective, education institutions should work toward developing the whole person. Of course, students need relevant education that enables them to earn a living and, at the same time, contribute to economic development. But this education should also contribute to a sense of social responsibility and self-fulfillment.

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